Math 526, S05 **Name:**

Quiz 9 **SSN: xxx-xx-**

<u>Instructions:</u> **There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$, find the eigenvalues and corrseponding eigenvectors of $\mathbf{A}$.

Solution:

$$\begin{aligned} det(\mathbf{A} - \lambda \mathbf{I}) &= 0 \\ det \begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix} &= 0 \\ (1-\lambda)(4-\lambda) - 10 &= 0 \\ \lambda^2 - 5\lambda - 6 &= 0 \\ (\lambda - 6)(\lambda + 1) &= 0 \\ \lambda &= 6, -1 \end{aligned}$$

For the eigenvalue $\lambda = 6$, we solve $(A - 6\mathbf{I})\mathbf{x} = \mathbf{0}$ to find the corresponding eigenvectors:

$$\begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_2 = \frac{5}{2} x_1$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ 5x_1/2 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 5/2 \end{bmatrix}$$

For the eigenvalue $\lambda = -1$, we solve $(A + \mathbf{I})\mathbf{x} = \mathbf{0}$ to find the corresponding eigenvectors:

$$\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_2 = -x_1$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ -x_1 \end{bmatrix} = c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$